Questions

Q1.

(i) Solve

$$7^{x+2} = 3$$

giving your answer in the form $x = \log_7 a$ where a is a rational number in its simplest form.

(3)

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(ii) Using the laws of logarithms, solve

$$1 + \log_2 y + \log_2 (y + 4) = \log_2 (5 - y)$$

(5)

(Total for question = 8 marks)

(Q03 WMA12/01, Jan 2021)

Q2.

The weight of a baby mammal is monitored over a 16-month period.

The weight of the mammal, w kg, is given by

$$w = \log_a(t + 5) - \log_a 4 \qquad 2 \leqslant t \leqslant 18$$

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where t is the age of the mammal in months and a is a constant.

Given that the weight of the mammal was 10 kg when $t = 3$

(a) show that $a = 1.072$ correct to 3 decimal places.

(3)

Using $a = 1.072$

(b) find an equation for t in terms of w

(3)

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(c) find the value of t when $w = 15$, giving your answer to 3 significant figures.

(2)

(Total for question = 8 marks)

(Q04 WMA12/01, Oct 2022)

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Q3.

Given $a = \log_2 3$

(i) write, in simplest form, in terms of a,

(a) $\log_2 9$

(b) $\log_2\left(\frac{\sqrt{3}}{16}\right)$

(3)

(ii) Solve

$$3^x \times 2^{x+4} = 6$$

giving your answer, in simplest form, in terms of a.

(4)

(Total for question = 7 marks)

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(Q10 WMA12/01, Oct 2022)

Q4.

(a) Sketch the curve with equation

$$y = 3 \times 4^x$$

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showing the coordinates of any points of intersection with the coordinate axes.

(2)

The curve with equation $y = 6^{1-x}$ meets the curve with equation $y = 3 \times 4^x$ at the point P.

(b) Show that the x coordinate of P is $\dfrac{\log_{10} 2}{\log_{10} 24}$

(5)

(Total for question = 7 marks)

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(Q09 WMA12/01, Jan 2020)

Q5.

(i) Find the exact value of x for which

$$\log_2(2x) = \log_2(5x + 4) - 3$$

(4)

(ii) Given that

$$\log_a y + 3\log_a 2 = 5$$

express y in terms of a. Give your answer in its simplest form.

(3)

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(Total for question = 7 marks)

(Q06 WMA12/01, Specimen papers)

Q6.

Use the laws of logarithms to solve

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$$\log^2(16x) + \log_2(x + 1) = 3 + \log_2(x + 6)$$

(Total for question = 5 marks)

(**Q05 WMA12/01, June 2023**)

Q7.

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In this question you must show all stages of your working.

Give your answers in fully simplified surd form.

Given that a and b are positive constants, solve the simultaneous equations

$$a - b = 8$$
$$\log_4 a + \log_4 b = 3$$

(Total for question = 6 marks)

(**Q04 WMA12/01, June 2022**)

Q8.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

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A software developer released an app to download.

The numbers of downloads of the app each month, in thousands, for the first three months after the app was released were

$$2k - 15 \qquad k \qquad k + 4$$

where k is a constant.

Given that the numbers of downloads each month are modelled as a geometric series,

(a) show that $k^2 - 7k - 60 = 0$

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(2)

(b) predict the number of downloads in the 4th month.

(4)

The **total** number of all downloads of the app is predicted to exceed 3 million for the first time in the Nth month.

(c) Calculate the value of N according to the model.

(Total for question = 6 marks)

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(**Q06 WMA12/01, June 2023**)

Q9.

The first term of a geometric series is 20 and the common ratio is $\frac{7}{8}$. The sum to infinity of the series is S_∞

(a) Find the value of S_∞

(2)

The sum to N terms of the series is S_N

(b) Find, to 1 decimal place, the value of S_{12}

(2)

(c) Find the smallest value of N, for which $S_\infty - S_N < 0.5$

(4)

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(Total for question = 8 marks)

(**Q02 WMA12/01, Specimen papers**)

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Q10.

Given $\log_a b = k$, find, in simplest form in terms of k,

(i) $\log_a\left(\frac{\sqrt{a}}{b}\right)$

(2)

(ii) $\frac{\log_a a^2 b}{\log_a b^3}$

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(2)

(iii) $\sum_{n=1}^{50}(k + \log_a b^n)$

(3)

(Total for question = 7 marks)

(**Q07 WMA12/01, Oct 2019**)

Q11.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

(i) Solve

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$$3^a = 70$$

giving the answer to 3 decimal places.

(2)

(ii) Find the exact value of b such that

$$4 + 3\log_3 b = \log_3 5b$$

(4)

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(Total for question = 6 marks)

(**Q05 WMA12/01, Oct 2023**)

Q12.

A scientist is using carbon-14 dating to determine the age of some wooden items.

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The equation for carbon-14 dating an item is given by

$$N = k\lambda^t$$

where

- N grams is the amount of carbon-14 **currently** present in the item
- k grams was the **initial** amount of carbon-14 present in the item
- t is the number of years since the item was made
- λ is a constant, with $0 < \lambda < 1$

(a) Sketch the graph of N against t for $k = 1$

(2)

Given that it takes 5700 years for the amount of carbon-14 to reduce to half its initial value,

(b) show that the value of the constant λ is 0.999878 to 6 decimal places.

(2)

Given that Item A

- is known to have had 15 grams of carbon-14 present initially
- is thought to be 3250 years old

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(c) calculate, to 3 significant figures, how much carbon-14 the equation predicts is currently in Item A.

(2)

Item B is known to have initially had 25 grams of carbon-14 present, but only 18 grams now remain.

(d) Use algebra to calculate the age of Item B to the nearest 100 years.

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(3)

(Total for question = 9 marks)

(Q09 WMA12/01, June 2022)

Q13.

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(i) Find the exact solution of the equation

$$8^{2x+1} = 6$$

giving your answer in the form $a + b \log_2 3$, where a and b are constants to be found.

(4)

(ii) Using the laws of logarithms, solve

$$\log_5(7 - 2y) = 2\log_5(y + 1) - 1$$

(5)

(Total for question = 9 marks)

(Q08 WMA12/01, June 2019)

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Q14.

(i) Find the exact value of x for which

$$\log_3(x + 5) - 4 = \log_3(2x - 1)$$

(4)

(ii) Given that

$$3^{y+3} \times 2^{1-2y} = 108$$

(a) show that

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$0.75^y = 2$

(4)

(b) Hence find the value of y, giving your answer to 3 decimal places.

(2)

(Total for question = 10 marks)

(**Q09 WMA12/01, Oct 2020**)

Q15.

Using the laws of logarithms, solve

$$\log_3 (32 - 12x) = 2 \log_3 (1 - x) + 3$$

(Total for question = 5 marks)

(**Q04 WMA12/01, Jan 2022**)

Q16.

A colony of bees is being studied.

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The number of bees in the colony at the start of the study was 30 000

Three years after the start of the study, the number of bees in the colony is 34 000

A model predicts that the number of bees in the colony will increase by p % each year, so that the number of bees in the colony at the end of each year of study forms a geometric sequence.

Assuming the model,

(a) find the value of p, giving your answer to 2 decimal places.

(3)

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According to the model, at the end of N years of study the number of bees in the colony exceeds 75 000

(b) Find, showing all steps in your working, the smallest integer value of N.

(5)

(Total for question = 8 marks)

(**Q05 WMA12/01, Jan 2020**)

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Q17.

A metal post is repeatedly hit in order to drive it into the ground.

Given that

- on the 1st hit, the post is driven 100 mm into the ground
- on the 2nd hit, the post is driven an **additional** 98 mm into the ground
- on the 3rd hit, the post is driven an **additional** 96 mm into the ground
- the **additional** distances the post travels on each subsequent hit form an arithmetic sequence

(a) show that the post is driven an **additional** 62 mm into the ground with the 20th hit.

(1)

(b) Find the **total distance** that the post has been driven into the ground after 20 hits.

(2)

Given that for each subsequent hit after the 20th hit

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- the **additional** distances the post travels form a geometric sequence with common ratio r
- on the 22nd hit, the post is driven an **additional** 60 mm into the ground

(c) find the value of r, giving your answer to 3 decimal places.

(2)

After a total of N hits, the post will have been driven more than 3 m into the ground.

(d) Find, showing all steps in your working, the smallest possible value of N.

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(4)

(Total for question = 9 marks)

(Q08 WMA12/01, Jan 2022)

Q18.

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(i) Using the laws of logarithms, solve

$$\log_3(4x) + 2 = \log_3(5x + 7)$$

(3)

(ii) Given that

$$\sum_{r=1}^{2} \log_a\left(y^r\right) = \sum_{r=1}^{2} \left(\log_a y\right)^r \qquad y > 1, a > 1, y \neq a$$

find y in terms of a, giving your answer in simplest form.

(3)

(Total for question = 6 marks)

(Q04 WMA12/01, Jan 2023)

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Q19.

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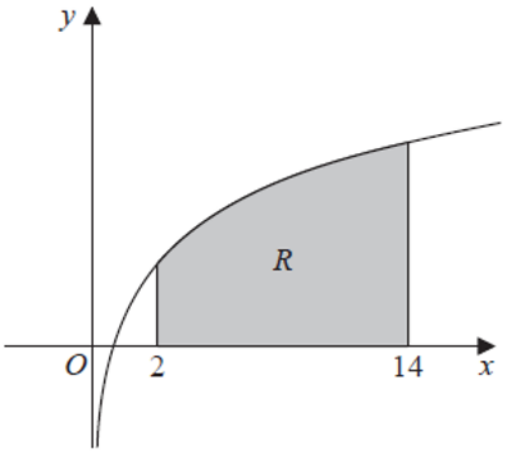


Figure 1

Figure 1 shows a sketch of part of the curve with equation $y = \log_{10} x$

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The region R, shown shaded in Figure 1, is bounded by the curve, the line with equation $x = 2$, the x-axis and the line with equation $x = 14$

Using the trapezium rule with four strips of equal width,

(a) show that the area of R is approximately 10.10

(3)

(b) Explain how the trapezium rule could be used to obtain a more accurate estimate for the area of R.

(1)

(c) Using the answer to part (a) and making your method clear, estimate the value of

(i) $\int_2^{14} \log_{10} \sqrt{x}\, dx$

(ii) $\int_2^{14} \log_{10} 100x^3\, dx$

(4)

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(Total for question = 8 marks)

(Q03 WMA12/01, Oct 2021)

Q20.

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The table below shows corresponding values of x and y for $y = \log_2(2x)$

The values of y are given to 2 decimal places as appropriate.

x	2	5	8	11	14
y	2	3.32	4	4.46	4.81

Using the trapezium rule with all the values of y in the given table,

(a) obtain an estimate for $\int_2^{14} \log_2(2x)\,dx$, giving your answer to one decimal place.

(3)

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Using your answer to part (a) and making your method clear, estimate

(b) (i) $\int_2^{14} \frac{\log_2(4x^2)}{5}\,dx$

(ii) $\int_2^{14} \log_2\left(\frac{2}{x}\right)dx$

(4)

(Total for question = 7 marks)

(Q01 WMA12/01, Jan 2020)

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